Computer Organization
And
Assembly Language

<u>Lab Manual 07</u>

Objectives:

1. To learn how shift and and rotate instructions work
2. To Print values of registers using shift/rotate and logic instructions
3. To understand and apply how can we use shift instructions to multiply and divide with power of 2.

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LAB TASK

NOTE: You are required to prepare the ASM file of the following task. Use maximum number of procedures to do the tasks.

NOTE: Read instructions carefully.

→ Use hard coded values to evaluate functions.

→ Use only shift instructions for multiplication and division purposes (You can also use add/sub(tract) instructions where required).

→ The procedures should accept first argument in bx/bl and second in dx/dl(if applicable).

→Use logic instructions for the first two procedures.

→ Use properly push and pop to avoid losing values of registers.

Write an assembly program which contains

- A procedure which accpets a number and checks whether it is even or odd.
- A procedure which accepts a character(ascii) and and checks whether it is capital or small.
- A procedure which accepts a number as parameter and multiplies it by 8.
- A procedure which accepts a number as parameter and multiplies it by 16.
- A procedure which accepts a number as parameter and multiplies it by 2.
- A procedure which accepts a number as parameter and multiplies it by 4.
- A procedure which accepts a number as parameter and divide it by 8.
- A procedure which accepts a number as parameter and divide it by 16.
- A procedure which accepts a number as parameter and divide it by 2.
- A procedure which accepts a number as parameter and divide it by 4.
- A procedure which accepts a number and a power and multiplies the given number by given power of 2.
- A procedure which accepts a number and a power and divides the given number by given power of 2.
- A procedure which accepts a 16 bit number and prints it in binary.
- A procedure which accepts a 16 bit number and prints it in hex.
- A procedure which accepts a 8 bit number and prints it in binary.
- A procedure which accepts a 8 bit number and prints it in hex.
- A procedure which accepts a number and multiplies it by 10.
- A procedure which accepts a number and multiplies it by 13.
- A main procedure which will pass arguments to the procedures and show their implementation.